

Q1.1

(c)
$$\begin{aligned} x + y + 3 &= 0 \\ -x - 2y + 7 &= 0 \end{aligned}$$

$$\tilde{l}_1 = \begin{bmatrix} 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x + y = -3$$

$$x + 2y = 7$$

$$\tilde{l}_2 = \begin{bmatrix} -1 & -2 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 7 \end{bmatrix} R_2 - R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 10 \end{bmatrix}$$

$$\begin{aligned} \tilde{x} &= \tilde{l}_1 \times \tilde{l}_2 \\ &= [\tilde{l}_1]_x \tilde{l}_2 \end{aligned}$$

NOTE:

$$[X]_x = \begin{bmatrix} 0 & z & -y \\ -z & 0 & x \\ y & -x & 0 \end{bmatrix}$$

$$y = 10$$

$$x + y = -3$$

$$x + 10 = -3$$

$$x = -13$$

$$\begin{bmatrix} -13 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & -1 \\ -3 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 7 \end{bmatrix}$$

$$\begin{aligned} 3(-2) - 1(7) \\ -3(-1) + 1(7) \\ 1(-1) - 1(-2) \end{aligned}$$

$$\begin{bmatrix} -13 \\ 10 \\ 1 \end{bmatrix}$$

getting non-homogenous coordinates:

$$x = -13, y = 10$$

So, yes we got the same point.

(d) $l_1 = (3, 4)$

(e) $\tilde{l}_1 = \begin{bmatrix} 2 \\ 5 \\ \frac{\sqrt{29}}{5} \end{bmatrix}$

$$\frac{1}{(\sqrt{3^2 + 4^2})} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2^2 + 5^2}} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{29}} \\ \frac{5}{\sqrt{29}} \\ \frac{1}{5} \end{bmatrix}$$

distance from origin is $\frac{1}{5}$

Q1.2

(a) $(1, 2)^T \rightarrow (0, 3)^T$

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

(b) $(x_i, y_i) = \left(\begin{bmatrix} x_1^i \\ x_2^i \end{bmatrix}, \begin{bmatrix} y_1^i \\ y_2^i \end{bmatrix} \right)$

$$E(T) = \sum_{i=1}^n \|Tx_i - y_i\|_2^2$$

$$T = \begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \end{bmatrix}$$

$$Tx_i = \begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \end{bmatrix} \begin{bmatrix} x_1^i \\ x_2^i \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1^i + t_1 \\ x_2^i + t_2 \end{bmatrix}$$

$$Tx_i - y_i = \begin{bmatrix} x_1^i + t_1 \\ x_2^i + t_2 \end{bmatrix} - \begin{bmatrix} y_1^i \\ y_2^i \end{bmatrix}$$

(c) $t_1 = \frac{1}{3} [(3-0) + (7-5) + (5-4)]$

$$= \frac{3+2+1}{3}$$

$$= 2$$

$$t_2 = \frac{1}{3} [(-5-1) + (6-7) + (-4-1)]$$

$$= \frac{-6-1-5}{3}$$

$$= \frac{-12}{3}$$

$$= -4$$

So,

$$= \begin{bmatrix} x_1^i + t_1 - y_1^i \\ x_2^i + t_2 - y_2^i \end{bmatrix} \quad [1/2]$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -4 \end{bmatrix}$$

Now,

$$E(T) = \sum_{i=1}^N (x_1^i + t_1 - y_1^i)^2 + (x_2^i + t_2 - y_2^i)^2$$

Now,

$$\frac{\partial E}{\partial t_1} = \sum_{i=1}^N -2(x_1^i + t_1 - y_1^i) + 0$$

$$= \sum_{i=1}^N 2(x_1^i - t_1 + y_1^i) = 0$$

$$= 2 \sum_{i=1}^N (x_1^i - t_1 + y_1^i) = 0$$

$$\sum_{i=1}^N (x_1^i - t_1 + y_1^i) = 0$$

$$\sum_{i=1}^N (y_1^i - x_1^i) - N t_1 = 0$$

$$t_1 = \frac{1}{N} \sum_{i=1}^N (y_1^i - x_1^i)$$

Similarly for t_2 :

$$t_2 = \frac{1}{N} \sum_{i=1}^N (y_2^i - x_2^i)$$

Q1.3

(a)

$$P = \begin{bmatrix} k & 0 \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} p & t \\ 0^T & 1 \end{bmatrix}$$

=

$$k = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 100 & 0 & 25 & 0 \\ 0 & 100 & 25 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & \cos 90^\circ & -\sin 90^\circ & 0 \\ 0 & \sin 90^\circ & \cos 90^\circ & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 100 & 0 & 25 & 0 \\ 0 & 100 & 25 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 100 & 25 & 0 & 150 \\ 0 & 25 & -100 & 50 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) $\tilde{x}_w = P^{-1} x_s$

$$= \begin{bmatrix} 1/100 & 0 & -1/4 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & -1/100 & 1/4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 25 \\ 50 \\ 1 \\ 0.25 \end{bmatrix}$$

$$= \begin{bmatrix} 0.25 \\ 0.5 \\ -0.25 \\ 0.25 \end{bmatrix}$$

$$= \frac{100}{25} \begin{bmatrix} 0.25 \\ 0.5 \\ -0.25 \\ 0.25 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -1 \\ 1 \end{bmatrix}$$

So, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

[7]

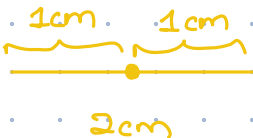
(C)

(i) C_0 centered at $c_c = (0, 0, 15)^T$ with equal sides $s=20$

$$\bar{x}_s = K \bar{x}_c \quad K = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

Corners of the cube are:

Explanation:

↳  , So each face can

be reached by moving $s/2$ in either direction.

Total combinations are:

$$2(\text{left-right } x) \cdot 2(\text{up-down } z) \cdot 2(\text{front-back } y) = 8$$

$$\bar{x}_{c_1} = \begin{bmatrix} -10 \\ -10 \\ 5 \end{bmatrix}$$

$$\bar{x}_{c_2} = \begin{bmatrix} 10 \\ -10 \\ 5 \end{bmatrix}$$

$$\bar{x}_{c_3} = \begin{bmatrix} 10 \\ 10 \\ 5 \end{bmatrix}$$

$$\bar{x}_{c_4} = \begin{bmatrix} -10 \\ 10 \\ 5 \end{bmatrix}$$

$$\bar{x}_{c_5} = \begin{bmatrix} -10 \\ -10 \\ 25 \end{bmatrix}$$

$$\bar{x}_{c_6} = \begin{bmatrix} 10 \\ -10 \\ 25 \end{bmatrix}$$

$$\bar{x}_{c_7} = \begin{bmatrix} 10 \\ 10 \\ 25 \end{bmatrix}$$

$$\bar{x}_{c_8} = \begin{bmatrix} -10 \\ 10 \\ 25 \end{bmatrix}$$

Now,

$$\bar{x}_{s_i} = K \bar{x}_{c_i} \quad i \in \{1-8\}$$

It would be better to write this as

$$X_{S_1} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ -10 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

$$\bar{X}_{S_2} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 5 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 5 \end{bmatrix}$$

$$\bar{X}_{S_3} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 100 \\ 100 \\ 5 \end{bmatrix}$$

$$\bar{X}_{S_4} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 100 \\ 5 \end{bmatrix}$$

$$\bar{X}_{S_5} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ -10 \\ 25 \end{bmatrix} = \begin{bmatrix} 200 \\ 200 \\ 25 \end{bmatrix}$$

$$\bar{X}_{S_6} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 25 \end{bmatrix} = \begin{bmatrix} 300 \\ 200 \\ 25 \end{bmatrix}$$

$$\bar{X}_{S_7} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 25 \end{bmatrix} = \begin{bmatrix} 300 \\ 300 \\ 25 \end{bmatrix}$$

$$X_{S_8} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -10 \\ 10 \\ 25 \end{bmatrix} = \begin{bmatrix} 200 \\ 300 \\ 25 \end{bmatrix}$$

S_0

$$\bar{X}_{S_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_2} = \begin{bmatrix} 20 \\ 0 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_3} = \begin{bmatrix} 20 \\ 20 \\ 1 \end{bmatrix}$$

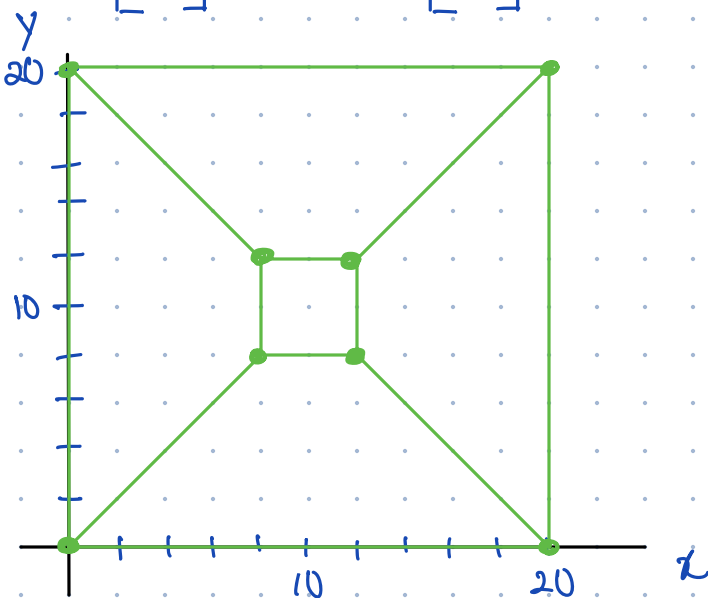
$$\bar{X}_{S_4} = \begin{bmatrix} 0 \\ 20 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_5} = \begin{bmatrix} 8 \\ 8 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_6} = \begin{bmatrix} 12 \\ 8 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_7} = \begin{bmatrix} 12 \\ 12 \\ 1 \end{bmatrix}$$

$$\bar{X}_{S_8} = \begin{bmatrix} 8 \\ 12 \\ 1 \end{bmatrix}$$



$$(iv) \bar{x}_s = K \bar{x}_c$$

$$\bar{x}_{s1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -10 \\ -10 \\ 5 \end{bmatrix} = \begin{bmatrix} -10 \\ -10 \end{bmatrix}$$

$$\bar{x}_{s2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ -10 \end{bmatrix}$$

$$\bar{x}_{s3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

$$\bar{x}_{s4} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -10 \\ 10 \\ 5 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \end{bmatrix}$$

$$\bar{x}_{s5} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -10 \\ -10 \\ 25 \end{bmatrix} = \begin{bmatrix} -10 \\ -10 \end{bmatrix}$$

$$\bar{x}_{s6} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 25 \end{bmatrix} = \begin{bmatrix} 10 \\ -10 \end{bmatrix}$$

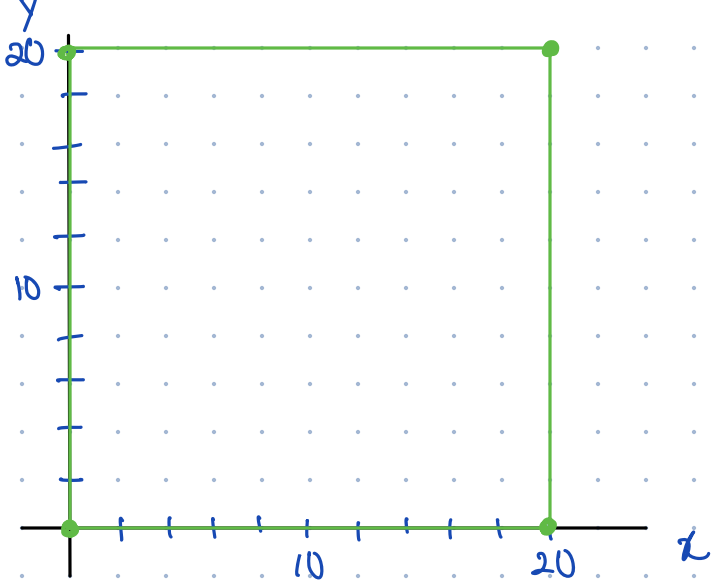
$$\bar{x}_{s7} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 25 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}$$

$$\bar{x}_{s8} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -10 \\ 10 \\ 25 \end{bmatrix} = \begin{bmatrix} -10 \\ 10 \end{bmatrix}$$

Now, add $(10, 10)^T$ to each.

$$\bar{x}_{s1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \bar{x}_{s2} = \begin{bmatrix} 20 \\ 0 \end{bmatrix} \quad \bar{x}_{s3} = \begin{bmatrix} 20 \\ 20 \end{bmatrix} \quad \bar{x}_{s4} = \begin{bmatrix} 0 \\ 20 \end{bmatrix} \quad \bar{x}_{s5} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\bar{x}_{s6} = \begin{bmatrix} 20 \\ 0 \end{bmatrix} \quad \bar{x}_{s7} = \begin{bmatrix} 20 \\ 20 \end{bmatrix} \quad \bar{x}_{s8} = \begin{bmatrix} 0 \\ 20 \end{bmatrix}$$



(v) At increasing focal lengths & distances

Q1.4

$$(a) \quad \frac{1}{f} = \frac{1}{z_s} + \frac{1}{z_c}$$

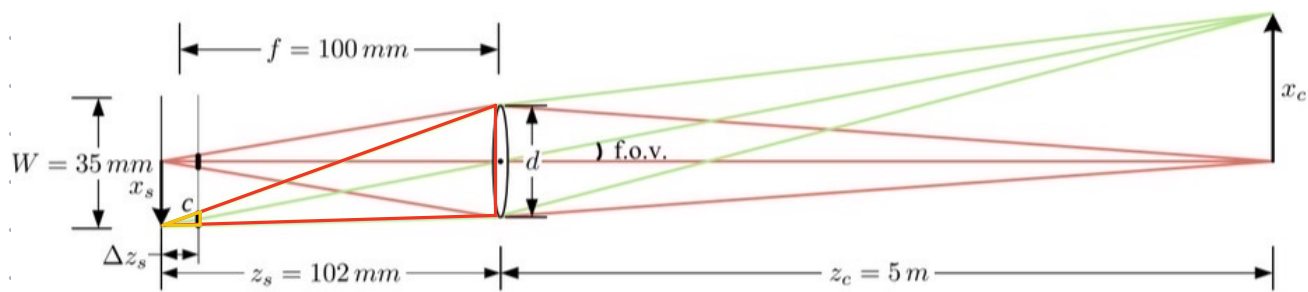
$$-\frac{1}{z_c} = \frac{1}{z_s} - \frac{1}{f}$$

$$= \frac{1}{104} - \frac{1}{100}$$

$$-\frac{1}{z_c} = -0.0038$$

$$z_c = 2599 \text{ mm}$$

(b)



From yellow & red triangles.

$$\frac{c}{\Delta z_s} = \frac{d}{z_s}$$

$$c = \frac{d}{z_s} \Delta z_s$$

$$\text{As } N = \frac{f}{d}$$

$$d = \frac{f}{N}$$

so,

$$c = \frac{f}{N z_s} \Delta z_s$$

(c)

$$c_1 = \frac{35 \text{ mm}}{1.4 \times 40 \text{ mm}} \times 0.1 \text{ mm} \quad c_2 = \frac{35 \text{ mm}}{1.4 \times 40 \text{ mm}} \times 0.03 \text{ mm}$$

$$= 0.0625 \text{ mm} \quad = 0.01875$$

An image is sharp, if size of $c <$ size of smallest pixel

$$\text{Pixel size} = \frac{\text{Sensor width}}{\text{Image Width}}$$

$$= \frac{8}{400}$$

$$= 0.02$$

So, second c_2 is in focus/sharp

Coding Exercise:

Q1.

(c)

$$L_{out}(p, v, \lambda) = L_{emit}(p, v, \lambda) + \int_{\Omega} BRDF(p, x, v, \lambda) \cdot \overset{1}{L_{in}}(p, x, \lambda) \cdot \overset{1}{(n^T r)} dx$$

$\hookrightarrow$ How much light is p emitting? $\hookrightarrow$ How light is reflected at point p for given x, v, λ ? $\hookrightarrow$ Incoming light

Affect of angle of L_{in} on amount of reflected light at p $\hookrightarrow N \parallel r$, max reflection
 $\hookrightarrow N \perp r$, 0

So,

$$L_{out}(p, v, \lambda) = \int_{\Omega} (n^T r) dx$$

